

Presentation Bundle

Optional Reference

These slides and posters were used for academic conference presentations and are provided for reference.

Conference Presentation Materials:

- EURO 2025 presentation slide deck
- Helmholtz AI spotlight presentation slide deck
- Helmholtz AI presentation poster

Dynamic Network (& AI) Analysis of Legal Citation Networks (especially medical) in Republic of Korea

2025. 6.

Chloe Kim*

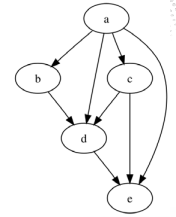
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Background

- **Judicial decision-making:** over 70 years of South Korean legal history represent a rich and evolving landscape.
- **Complex interlinked systems:** legal rulings do not exist in isolation—citations make them a form of complex systems.
- **Especially in medical law:** precedents shape institutional responsibilities, patients' rights, and norms of medical accountability.

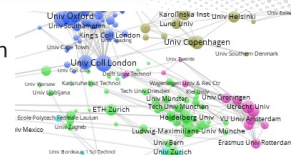


Research Gap

- **Legal networks in Korean judicial context:** while networks are widely studied in academia, legal citation networks remain underexplored.
- **Understanding structural evolution:** previous studies have focused on text mining and precedent frequency, but influential case dynamics (over time) could improve our awareness in (1) **psychological**, (2) **linguistic** and (3) **societal** perspectives.

Motivations

- **Structure of legal citations:** how it has been structured in a view of **networks**?
- **Hub nodes:** what are the key law precedents in different political periods (esp. in medical laws)?
- **Influential cases:** what are changes in influential cases (inequal cases) relevant to medical cases (compared to overall ones)?

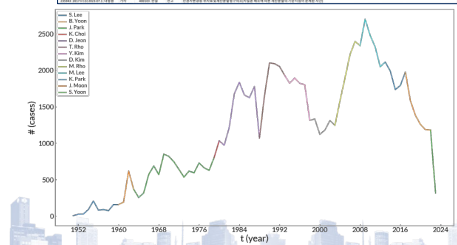
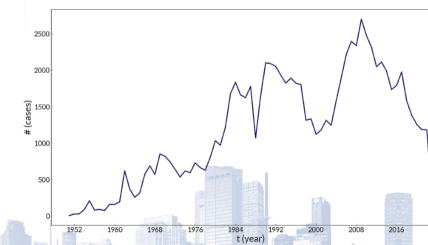


Approaches

- **Complex networks:** 86,000+ Korean legal cases (1951–2023, supreme)
 - Individual cases: nodes
 - Citation relationships: edges
- **Focused subnetwork:** medical-related cases for deeper domain-specific insight
- **Heavy-tails and hub:** non-Gaussian analysis, graph-theoretic centrality, presidential-cycle segmentation, and MDS optimization



- **Legal Cases:** 86,000+ Korean legal cases (1951–2023, supreme)
 - Information: case name, number, date of judgement, statement, case type, judgement type (supreme)
 - Citation: a list of **referred cases**

[illegible]



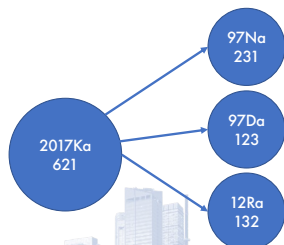
Target data -- Legal cases in Korea

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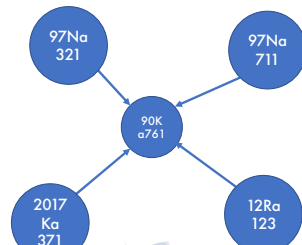
Complex networks

- individual cases: 86,161 nodes
- citation relationships: edges (from referred/referring)

Referring network



Referred network



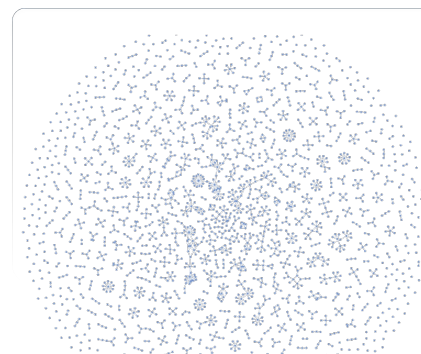
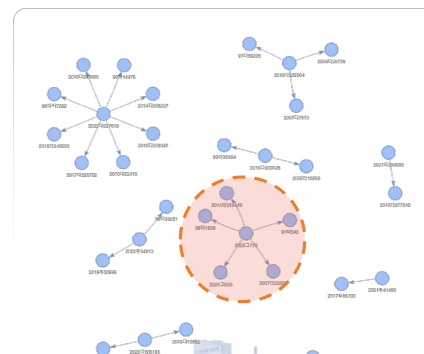
5



Comprehensive legal case citation networks

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Referring network



- Out-degree analysis: max 43 ~ min 1
- 2003Do3516(43) > 2004Da37775(32) > ...

- Nodes: 58,555 (w/o null-edge case)
- Edges: 115,790

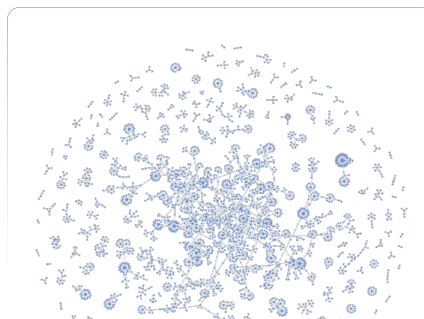
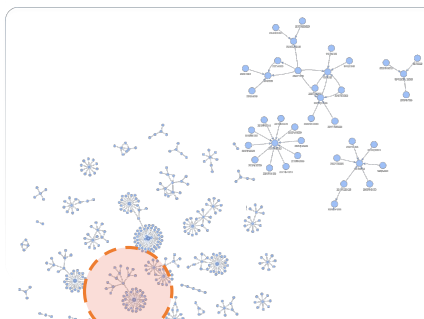
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Comprehensive legal case citation networks

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Referred network



- In-degree analysis: max 66 ~ min 1
- Hub-like: 95Da28625(66) > 90Da12243(47) > ...

- Nodes: 58,555 (w/o null-edge case)
- Edges: 115,790

- Hub-like both in referring & referred: 99Do4923, 90Do646, 97Do2609.

7

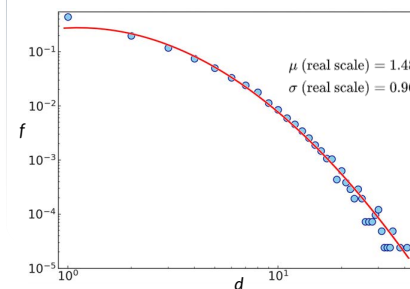


Comprehensive legal case citation networks

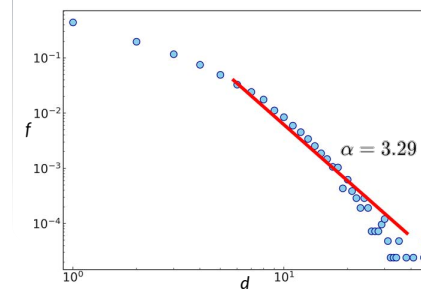
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Non-Gaussian analysis: Referred network

Body: lognormal

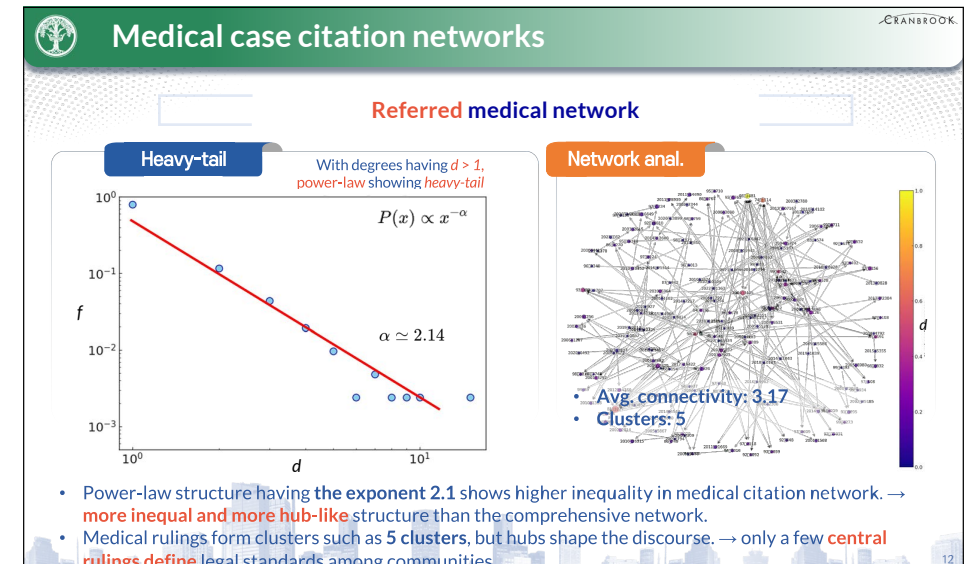
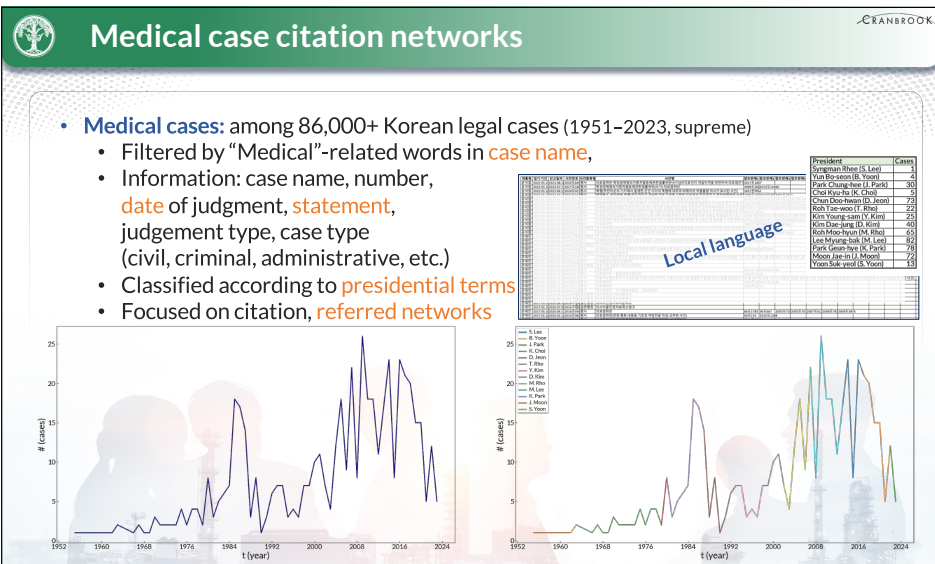
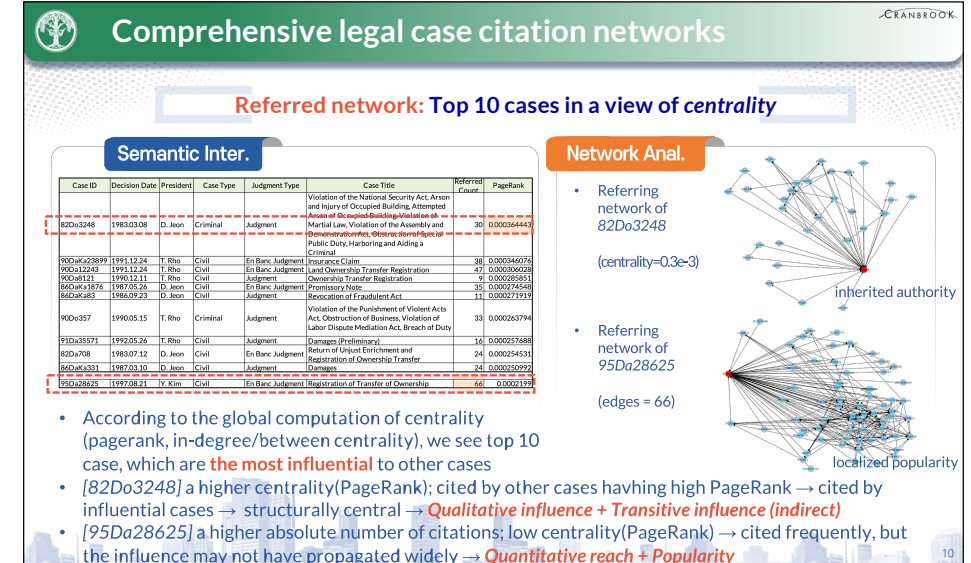
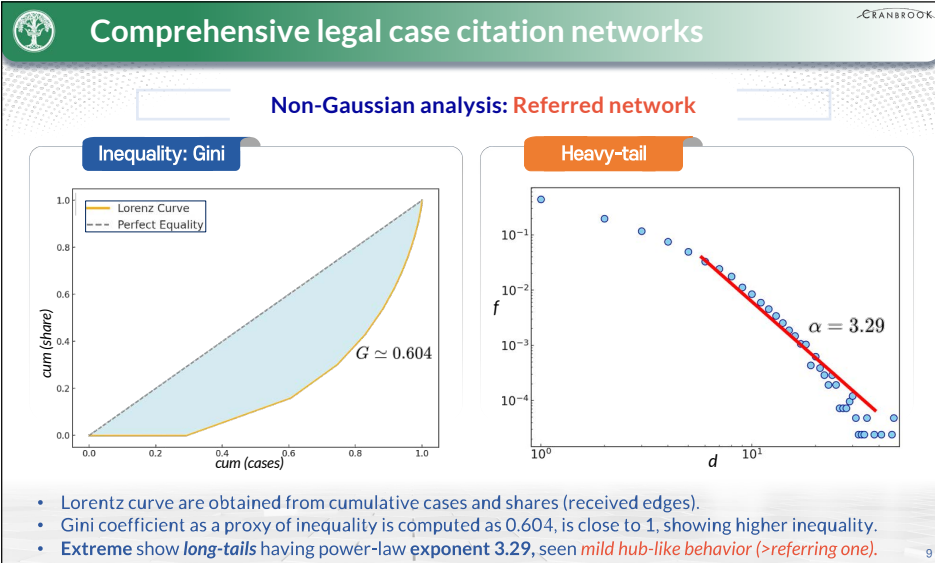


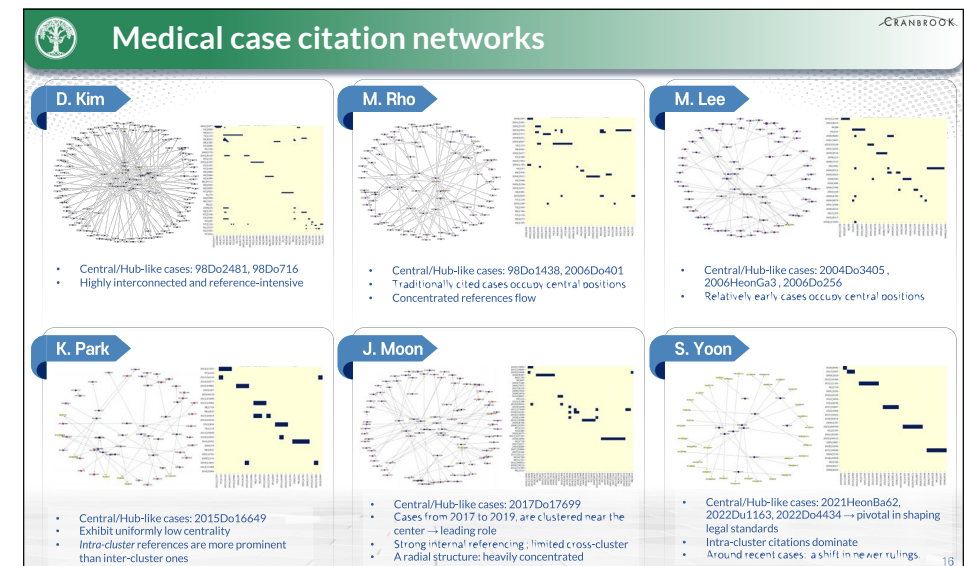
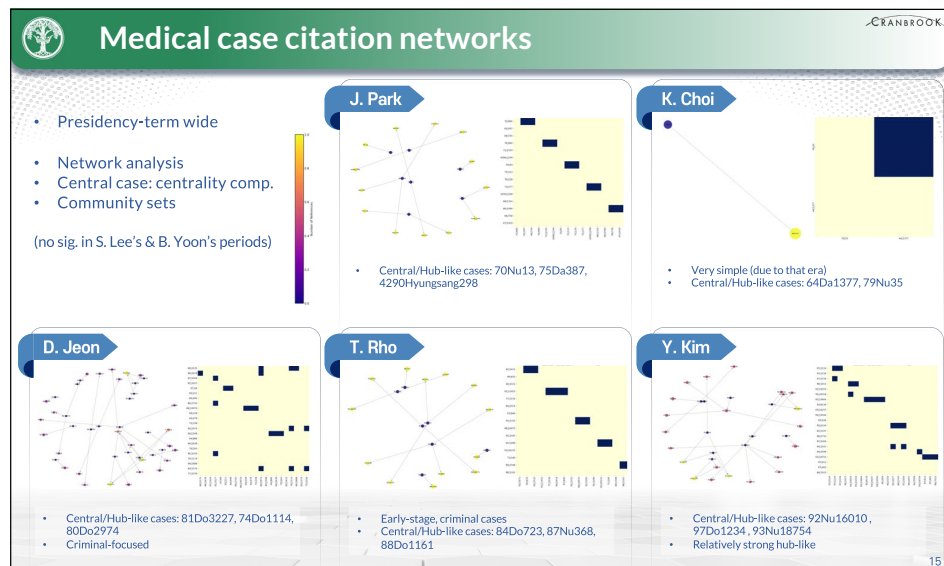
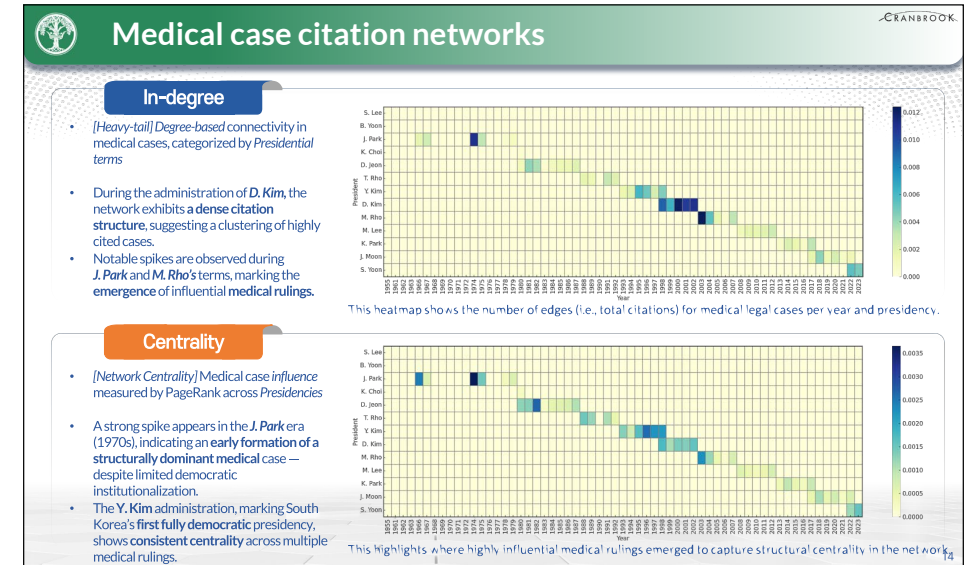
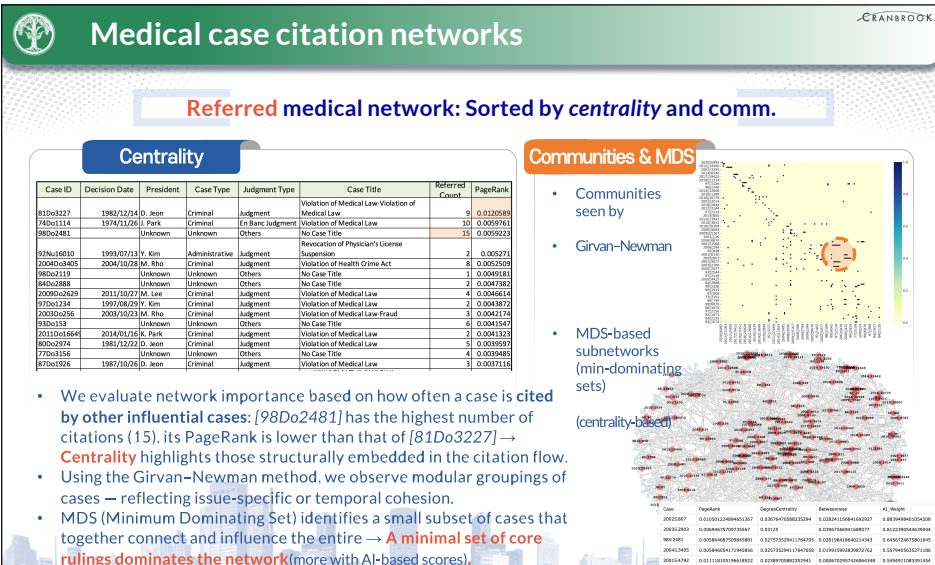
Tail: Heavy-tail



- The x-axis means the degree (number of edges) d and the y-axis for normalized frequency f .
- Body is distributed **left-skewed**, having expectations of 1.48.
- Extreme show **long-tails** having power-law exponent 3.29, seen **mild hub-like behavior** (>referring one).

8





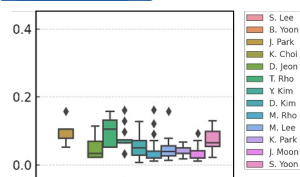


Medical case citation networks

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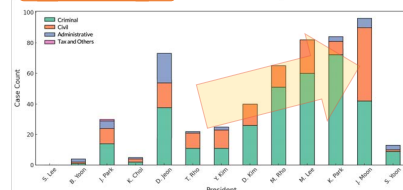
Semantic analysis in medical network

Centrality



- MDS-based centrality scores
- Presidency-wide
- Certain administrations --particularly T. Rho's and J. Moon's-- shows higher MDS-based centrality distributions indicating **more structurally central medical rulings** → **concentrated medical legal influence**.
- The legal nature of influential medical rulings shifted over time—from criminal liability in earlier regimes to civil and administrative disputes in later democratic periods.

Semantic analysis



- Analyzing hubs and central medical cases via reading and understanding the '**judgement statements**'
- Important legal words are **different in each period!**

17



Summary and Concluding remarks

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- **[Legal cases in Rep. of Korea]** 86,000+ Korean legal cases (1951–2023, supreme) are documented with semantic information such as case name, date, judgement statement, case type, and citations among them.
 - We classify them according to presidential terms and focus on citation.
- **[Structure of legal citations]** Power-law patterns indicate **hub-like (heavy-tailed) behavior** with scale-free characteristic, and they suggest hub- and central legal case nodes (referred & referring networks).
- **[Influential cases]** In a view of centrality and MDS, it is possible to identify the important & influential legal cases (*Qualitative influence & Quantitative popularity*).
- It provides practical implications for legal and policy decision-making in governance.



Summary and Concluding remarks

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- South Korea's **medical legal network** reveals a structural and semantic shift.
 - 1) Lower power-law exponent (~2.1) than overall cases network shows **higher inequality and hub-like behavior**.
 - 2) A structural concentration of centrality is shown (by centrality).
 - 3) Over time, there is **semantic shift** from **criminal** to **civil & administrative domains** (from centralized criminal rulings to broader civil and administrative influence).
 - 4) It reflecting **evolving norms** in healthcare governance.
- Structural and semantic dual analysis offers insights: policy design, legal reform, healthcare government strategy.

Thank you!

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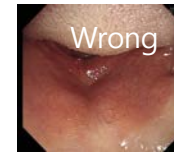
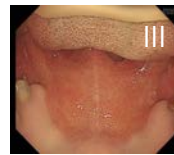
Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction

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- ✓ **Mallampati score** is simple but suffers from subjectivity and poor reproducibility.
 - Decisions like endoscopic assistance rely on clinical intuition → *lack of standardization*
- ✓ **AI** (Deep Learning & ViT) shows potential in airway assessment and intubation prediction.
- ✓ AI-powered Mallampati scoring can be improved.
 - Multimodal data integration: images + clinical variables
 - Increase in interpretability

- Class I
 - visualization of the soft palate, fauces, uvula, and both anterior and posterior pillars
- Class II
 - visualization of the soft palate, fauces, and uvula
- Class III
 - visualization of the soft palate and the base of the uvula
- Class IV
 - soft palate is not visible at all

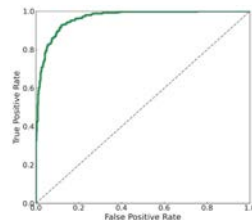


- ✓ De-identified data (image-based and structured) in a tertiary hospital (total 500)
 - **Mallampati photographs** (labeled)
 - **Features**
 - Score/Test
 - Age
 - BMI
 - Gender
 - Life habits

- Mallampati classification (4 cls)
- Endoscopy necessity/possibility
- **Confusion Mtx.**
- An LLM API-level prototype module (ad-hoc) supports *interpretability*.
- **AUROC**

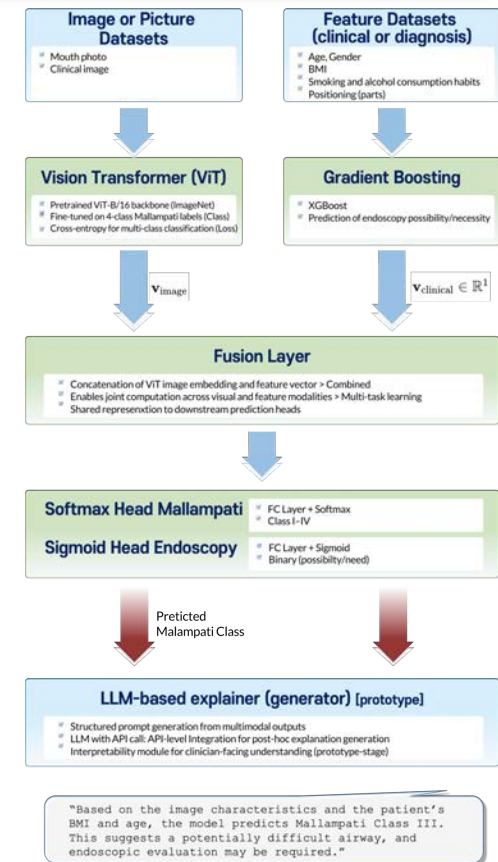
True Label \ Pred Label	Pred I	Pred II	Pred III	Pred IV
True I	112	6	3	2
True II	1	132	2	3
True III	2	2	117	7
True IV	1	3	4	103

- Classification accuracy of 92.7%
- 4-class softmax
- Robustness in diagonal



- Endoscopy possibility (0.9)
- sigmoid
- Discriminative capacity in decision

- ✓ Multimodal learning enables standardized, accurate Mallampati prediction & endoscopy possibility.
- ✓ Study highlights potential for scalable, clinically-relevant AI in airway evaluation.



Thank you!



An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI

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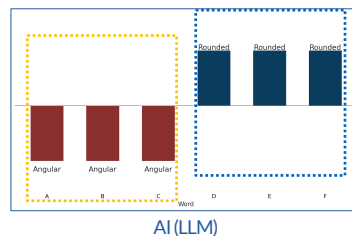
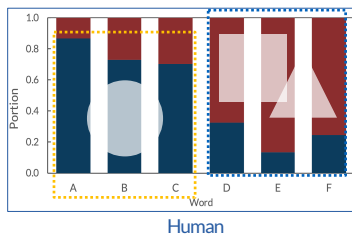


Introduction & Motivation

- ✓ **Perception:** Human intuition and LLM reasoning (may) differ fundamentally in multimodal perception.
 - Here it is focused on converting meaningless structured sounds into visual forms.
- ✓ **Motivation:** Understand differences in cognitive (intuitive) vs. statistical-semantic (LLM) processing.
 - Cognitive linguistics and sound symbolism (e.g., bouba-kiki effect) showed consistent mappings between phonetic forms and shapes in humans.
 - Recent AI models attempt to emulate human-like perception, their semantic reasoning remains distinct.
 - Explore limitations of human-like generative AI: not only to evaluate the capabilities, but to uncover the foundational principles of perception.
 - Motivated by these gaps, a comparative study is designed to isolate the mechanisms in humans and contrast them with LLMs' operations.
- ✓ **Medium:** Non-symbolic sound words serve as stimuli bridging auditory and visual modalities.

Approaches & Results

- ✓ **Collected:** the form of *shapes* (human) and *generated outputs* (AI).
 - Participants are instructed to visualize the sounds (A, B, C, D, E, F) and interpret.
 - Results were categorized into simple geometric classes: (1) Rounded (●/circle-like, smooth, curved) or (2) Angular (▲, ■/pointed, straight-edged).
 - Their interpretations are compared: against LLM-generated images based on those inputs.
- ✓ Distinctive results between the two -- our findings show a *clear divergence*.



- 1) *Humans relied on intuitive mapping:* **direct sensory-affective** visualizing the stimuli through curves or sharp (pointed) shapes intuitively.
- 2) *AI (LLM, GPT) used semantic reasoning:* mapping phonological cues to meanings from text, such as interpreting the sound into a specific shape **based on semantic associations learned** from language data.

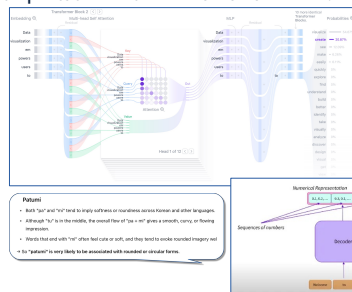
- ✓ Quantitative Evaluation, mismatch in shape interpretation between human and LLM.

- $R_i \in [0, 1]$: the proportion of participants categorized word i as Rounded.
- $G_i \in [0, 1]$: LLM's binary judgment.

$$D = \frac{1}{N} \sum_{i=1}^N |R_i - G_i| \approx 0.766$$

- ✓ Supports possibly a dual-mode hypothesis of cognition in human vs. AI processing.

- **Human:** driven by intuition without explicit prior learning
- **AI(LLM):** learned, symbolic reasoning.

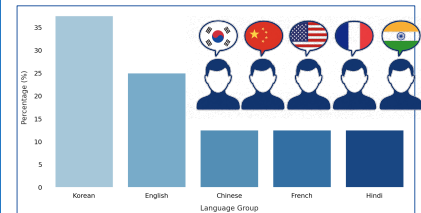


Targets and Data

Target Groups

We target two primary agents:

- ✓ Human participants (n ≈ 40) across Korean, Chinese, English, French, Hindi speakers.
- ✓ GPT-4 language model.



Stimuli

- ✓ Non-symbolic sound word
- ✓ **Auditory items**
 - lack semantic content
 - contain phonological structure.
- ✓ ['ʌn.gu:.ɪ.gʌŋ | 'æŋ.gəl.zər | 'kʲʌŋ.bək.ku:.neɪ | 'pou.kə.rə | 'pə:.tu:.mi: | 'gæl.u:.pi:] (anon. names A, B, C, D, E, F)

Task

- ✓ Visualize or generate shapes based on auditory input.

Summary

- ✓ This study investigates how humans and AI interpret non-symbolic sound stimuli through visual reasoning. The experimental approach tries to compare roundedness judgments across six abstract sound words. Findings reveal a clear cognitive gap between human visual generation and LLM-generated responses:
 - **Human perception** operates pre-linguistically, driven by affective intuition.
 - **LLM** reflects learned, language-anchored cognition, through corpus-based semantic inference.
- ✓ These contrasts underscore fundamental differences in generative mechanisms—highlighting directions for AI with human-like perceptual grounding.

References

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Thank you!

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